

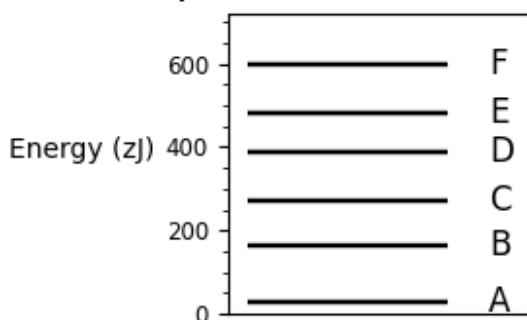
Name: _____

These questions assess your understanding of the material from Chapters 7, 8 of Silberberg. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must explain why you chose your answer. Remember to report the units and correct number of significant figures in your answers.

1. Imagine an alternate universe in which the value of the Planck constant is $5.63 \times 10^{-32} \text{ J}\cdot\text{s}$. In that universe, is quantum mechanics necessary to accurately describe a 5.2×10^{-5} -m-wide object that has a mass of 1.6×10^{-27} kg and is moving at 5.2×10^{-6} m/s? Write "yes" if quantum mechanics is necessary and "no" if it is not, then explain why you made your choice. Answers lacking supporting explanation will earn no credit.

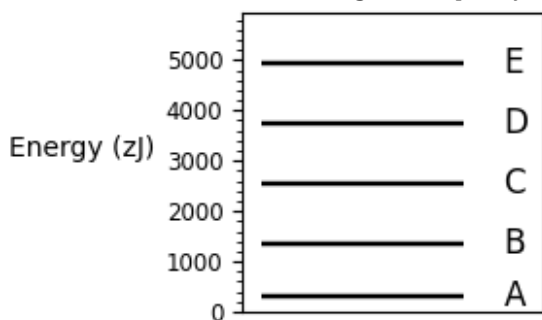
ANSWER: _____

2. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the absorption line at the shortest wavelength?



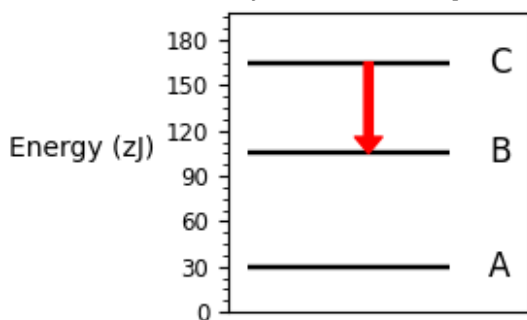
ANSWER: _____

3. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the emission line at the highest frequency?



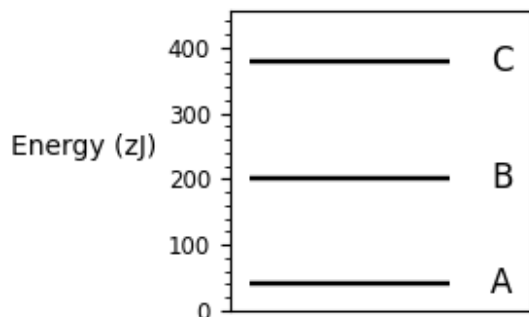
ANSWER: _____

4. This energy diagram shows the allowed energy levels of an electron in a certain atom. If the electron makes a transition from C to B, as shown by the arrow, will a photon be absorbed or emitted?



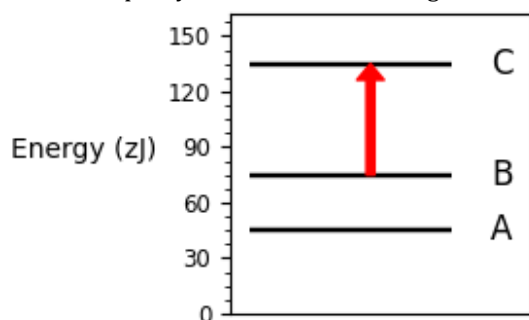
ANSWER: _____

5. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the emission line at the lowest frequency?



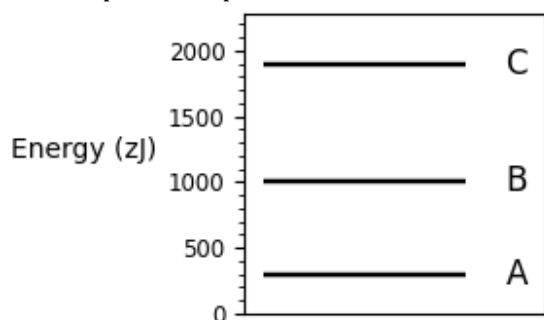
ANSWER: _____

6. This energy diagram shows the allowed energy levels of an electron in a certain atom. Suppose the electron makes a transition from B to C, as shown by the arrow. Calculate the wavelength of the photon that would be absorbed or emitted. Report your answer to three significant figures. Note that the SI prefix zepto (z) means 10^{-21} .



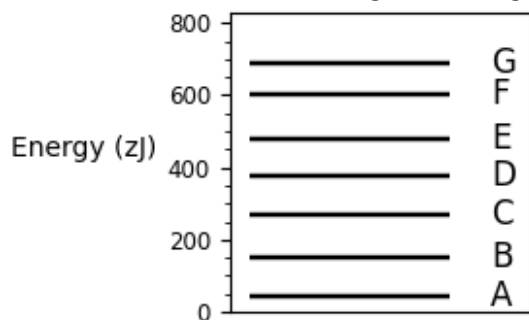
ANSWER: _____

7. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. How many lines are in the absorption line spectrum?



ANSWER: _____

8. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the emission line at the longest wavelength?



ANSWER: _____

9. List the following electron subshells in order of increasing energy. You may assume that these subshells are in an atom with many electrons.

4s, 4d, 5s, 2s, 3s

ANSWER: _____

10. Calculate the wavelength of the line in the emission line spectrum of hydrogen caused by the transition of the electron from an orbital with $n = 6$ to an orbital with $n = 2$. Round your answer to 3 significant figures.

ANSWER: _____

11. List the following electron subshells in order of increasing energy. You may assume that these subshells are in an atom with many electrons.

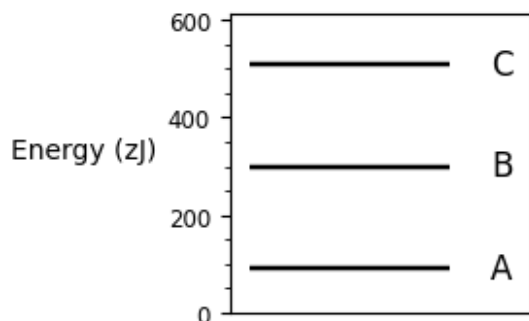
4s, 4d, 5s, 2s, 3s

ANSWER: _____

12. Several types of electromagnetic radiation (EMR) are: radio waves, green light, microwaves, and orange light. Order these types of EMR from lowest frequency to highest frequency.

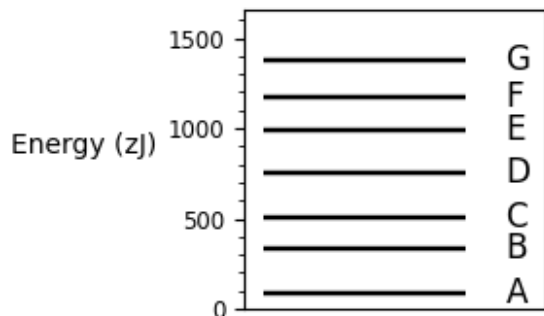
ANSWER: _____

13. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the absorption line at the longest wavelength?



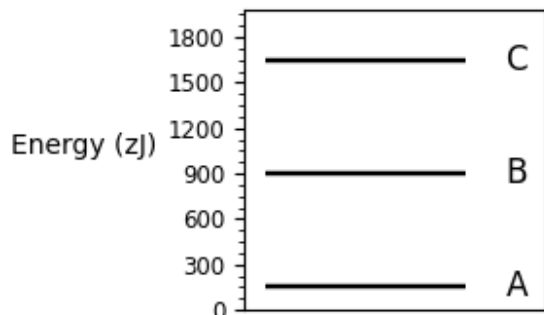
ANSWER: _____

14. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. How many lines are in the emission line spectrum?



ANSWER: _____

15. This energy diagram shows the allowed energy levels of an electron in a certain atom. What is the energy of the electron in its ground state? Note that the SI prefix zepto (z) means 10^{-21} .

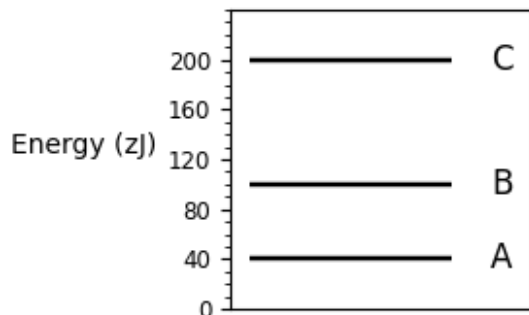


ANSWER: _____

16. An atom of a particular element has a radius of 110 pm and the average orbital speed of the electrons in it is approximately 8.32×10^7 m/s. Calculate the least possible uncertainty in a measurement of the speed of an electron in this atom. Write your answer as a percentage of the average speed and round it to two significant figures.

ANSWER: _____

17. This energy diagram shows the allowed energy levels of an electron in a certain atom. What is the energy of the electron in its second excited state? Note that the SI prefix zepto (z) means 10^{-21} .

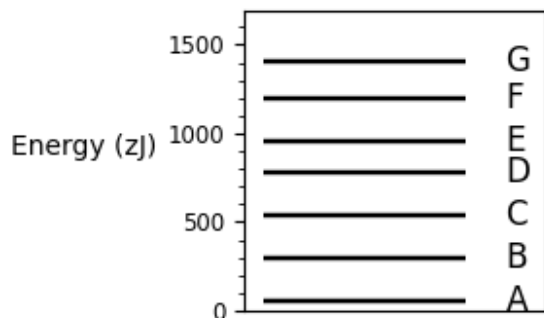


ANSWER: _____

18. Suppose a popular FM-radio station broadcasts radio waves with a frequency of 92.41 MHz. Calculate the wavelength of these radio waves.

ANSWER: _____

19. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the absorption line at the highest frequency?



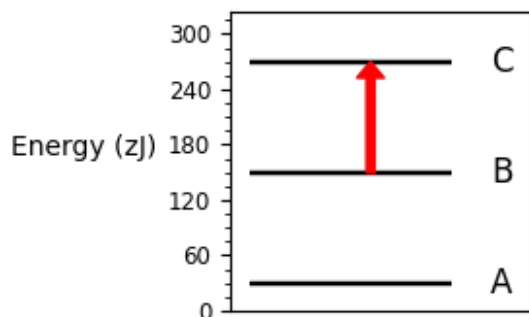
ANSWER: _____

20. Fill in the information missing from the following table. For each subshell, you must provide the principal quantum number n , the angular momentum quantum number l , and the maximum number of electrons that can fit in the subshell N_e . Credit is awarded only if the table is completed correctly.

Subshell	n	l	N_e
4p			
3p			
4d			

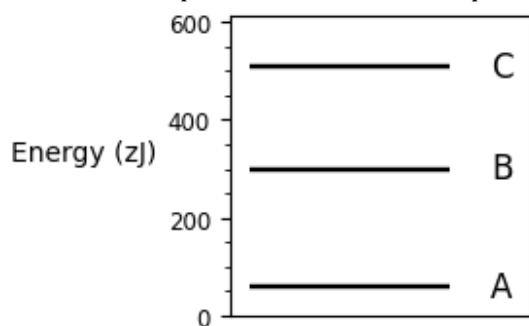
ANSWER: _____

21. This energy diagram shows the allowed energy levels of an electron in a certain atom. Suppose the electron makes a transition from B to C, as shown by the arrow. Calculate the energy of the photon that would be absorbed or emitted. Note that the SI prefix zepto (z) means 10^{-21} .



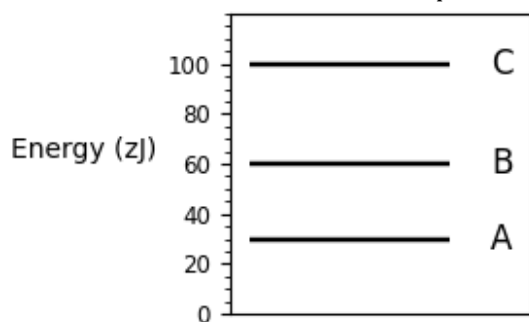
ANSWER: _____

22. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the absorption line at the lowest frequency?



ANSWER: _____

23. This energy diagram shows the allowed energy levels of an electron in a certain atom. What is the energy of the electron in its first excited state? Note that the SI prefix zepto (z) means 10^{-21} .

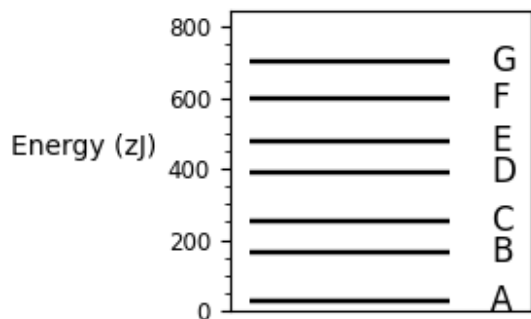


ANSWER: _____

24. Several types of electromagnetic radiation (EMR) are: red light, ultraviolet light, gamma rays, and green light. Order these types of EMR from lowest energy to highest energy.

ANSWER: _____

25. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the emission line at the shortest wavelength?



ANSWER: _____

26. Several types of electromagnetic radiation (EMR) are: infrared light, red light, x-rays, and blue light. Order these types of EMR from shortest wavelength to longest wavelength.

ANSWER: _____

27. It takes 199.2 kJ/mol to break a bond between two particular elements. Calculate the maximum wavelength of light that will break one of these bonds.

ANSWER: _____